

StickFrame — Technology Intelligence Report

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Prepared for Claude Code (Engineering Lead) & Codex CLI (Core Engine Lead)

1. Executive Summary

This report maps the open-source ecosystem for building **StickFrame** — a text-to-stickman-animation engine. The goal is to identify libraries, repos, and skills that reduce the amount of code Claude Code and Codex need to write from scratch.

Key Finding: You can save ~60% of boilerplate code by leveraging existing libraries. The core unique value (stickman rig + animation data model) must still be built from scratch.

Top 5 Quick Wins

#	Library	What It Replaces	Effort Saved
1	pytweening	Easing / tweening functions (30+ curves)	~1 day of math
2	movis	Video composition, layers, effects, ffmpeg export	~1 week of pipeline
3	Stickman-Studio	Reference architecture for stickman pipeline	Reference only
4	The-StickMan	Joint rotation math, stickman drawing code	~2 days of math
5	claudecodesignskills	Claude Code skills for 3D/animation/graphics	Skill scaffolding

2. Directly Reusable Libraries

2.1 Rendering Libraries

Library	Stars	License	Verdict for StickFrame
Pillow (PIL) github.com/python-pillow/Pillow	12k+	Historical	USE Already planned. Does line/circle drawing, compositing, GIF export.
pycairo github.com/pygobject/pycairo	1k+	LGPL	OPTION Better line quality (antialiased), vector-based. Could replace Pillow for drawing.
drawsvg github.com/cduck/drawsvg	600+	MIT	OPTION SVG generation → convert to PNG via cairo. Resolution independent.
pygame github.com/pygame/pygame	7k+	LGPL	OVERKILL Full game engine. Good for headless rendering but heavier than needed.

2.2 Animation & Easing Libraries

Library	Description	Verdict
pytweening github.com/asweigart/pytweening	30+ easing functions: linear, easeInQuad, easeOutCubic, easeInOutSine, easeInBounce, easeOutElastic, etc. Input 0.0-1.0 → output 0.0-1.0. Pure Python, zero deps.	DEFINITELY USE <code>pip install pytweening</code>
movis github.com/rezoo/movis	Python video editing engine: compositions, layers, keyframes, easing, blending modes, drop shadow, blur, chromakey, nested comps, ffmpeg export. MIT license.	STRONG CANDIDATE Could serve as entire rendering pipeline. Last commit 2 years ago — evaluate first.

Key Insight on movis: Movis has a `keypoint` and easing-based animation engine built-in. We could use movis *compositions* as our scene graph, with `add_layer()` for characters and objects, and let movis handle frame-by-frame rendering + ffmpeg export. We'd only need to write the custom stickman drawing layer (a function that returns RGBA numpy arrays per frame).

Install: `pip install movis`

2.3 Video Export Libraries

Library	Description	Verdict
ffmpeg-python github.com/kkroening/ffmpeg-python	Python bindings for ffmpeg with filter graph support.	USE IF NOT USING MOVIS
moviepy github.com/Zulko/moviepy	Classic Python video editing. Mature but heavier.	FALLBACK
FFmpeg (native) ffmpeg.org	Use via <code>subprocess</code> — no extra Python dep.	SIMPLEST Save PNG frames → ffmpeg stitch. Works every time.

2.4 Pre-Built Stickman Projects (Reference & Reuse)

Repo	Stars	What It Does	Reuse Potential
Stickman-Studio github.com/saiedpod-bot/Stickman-Studio	~50	AI-powered stickman animation video pipeline. Uses Gemini, Imagen, edge-tts, ffmpeg. Streamlit dashboard. Script → stickman video → YouTube.	REFERENCE ARCHITECTURE Not directly reusable (uses AI), but excellent reference for pipeline design. Study their slideshow mode + Ken Burns effect + ffmpeg stitching.
The-StickMan github.com/francisA88/The-StickMan	2	Stickman animation framework made with Kivy. Has RotateableLine, joint rotation, StickMan widget class.	MATH REFERENCE Small but has working joint rotation math. The <code>stickman.py</code> file has angle-based limb transformations we can reference. MIT license.
StickMotion github.com/InvertedForest/StickMotion	~100	Diffusion-based stickman motion generation from text + stickman conditions.	OVERKILL FOR MVP Uses diffusion models. Good for Year 3 AI layer, not for deterministic engine.
	~30		REFERENCE Good for understanding how to

DanceStickFigure github.com/aaryan-athena/DanceStickFigure		MediaPipe pose estimation → stickman animation. Streamlit app.	map joint coordinates to stickman drawings. Uses Pillow for rendering.
bone-animator github.com/sanskarchand/bone-animator	~10	Pivot Animator clone — manual stickman animation tool.	CONCEPT REFERENCE Shows how Pivot-style stickman animation works conceptually.
StickyPy stickypy.sourceforge.net	N/A	Professional stick figure animation program written in Python + Pygame. Pivot-like.	CONCEPT REFERENCE Old but shows complete stickman animation software built in Python.
Axis github.com/davepagurek/Axis	~100	Stick figure animator for keyframed animation.	REFERENCE Keyframed animation approach for stick figures. Study their interpolation model.

3. Recommended Architecture (Based on Research)

Recommendation: Use **Pillow + pytweneing** for the MVP (simplest path), evaluate **movis** for the production pipeline. Do NOT use game engines (pygame/arcade) — they're designed for interactivity, not headless frame rendering.

Option A: Pillow + pytweneing (MVP — Recommended)

```
sources/engine/
├─ scene.py           # Scene data models (Pydantic)
├─ animation.py       # AnimationPlayer + keyframe interpolation
├─ animation_clips.py # Walk/run/jump/attack/fall keyframe data
├─ rig.py             # Stickman skeleton joint resolver
├─ renderer.py        # Pillow-based frame renderer
├─ ffmpeg_export.py   # PNG sequence → ffmpeg → MP4
├─ __main__.py        # CLI entry point
└─ demo_scene.json    # Test scene
```

Dependencies: `pip install pillow pytweneing` + ffmpeg on PATH

Option B: movis Pipeline (Production — Strong Candidate)

```
sources/engine/
├─ scene.py           # Scene data models
├─ animation.py       # AnimationPlayer
├─ animation_clips.py # Keyframe data
├─ rig.py             # Skeleton joint resolver
├─ layers.py          # Movis-compatible custom layers (stickman drawing)
└─ __main__.py        # Builds movis Composition + adds layers → write_video()
```

Dependencies: `pip install movis pytweneing`

movis Risk: Last commit was 2 years ago (2024). The repo may be unmaintained. Test compatibility with Python 3.13 before committing to this path.

3.1 Stickman Skeleton Model (Based on The-StickMan + Axis Research)

Based on analysis of existing stickman frameworks, our joint hierarchy should be:

```
HEAD (center)           # Circle
|
BODY_TOP (neck/shoulders) # Origin for arms
|
BODY_BOTTOM (hips)       # Origin for legs

LEFT_SHOULDER — LEFT_ELBOW — LEFT_HAND    # Upper arm → forearm → hand
RIGHT_SHOULDER — RIGHT_ELBOW — RIGHT_HAND
LEFT_HIP — LEFT_KNEE — LEFT_FOOT           # Thigh → shin → foot
RIGHT_HIP — RIGHT_KNEE — RIGHT_FOOT
```

Each limb has a base angle + animation angle offset, as demonstrated in The-StickMan's `RotateableLine` class.

3.2 Keyframe Data Format (Based on Axis + Research)

Similar to Axis's approach but simpler — per-frame joint angle arrays with linear interpolation:

```
AnimationClip "walk":
  Frame 0: { left_arm: 30, right_arm: -30, left_leg: -25, right_leg: 25, body_tilt: 3 }
  Frame 2: { left_arm: 0, right_arm: 0, left_leg: 0, right_leg: 0, body_tilt: 0 }
  Frame 4: { left_arm: -30, right_arm: 30, left_leg: 25, right_leg: -25, body_tilt: -3 }
```

```
Frame 6: { left_arm: 0, right_arm: 0, left_leg: 0, right_leg: 0, body_tilt: 0 }  
Interpolation: linear (or use pytweneing.easeInOutSine for smoother motion)
```

4. Claude Code Skills for Animation/Graphics

Skill	Source	What It Does	Install
claudedesignskills	github.com/freshtechbro/claudedesignskills	3D graphics, animation, WebGL, Three.js, GSAP. 50+ skills for web development with animation focus. Plugin-based marketplace.	<code>/plugin marketplace add freshtechbro/claudedesignskills</code>
animation-principles	crossaitools.com	Disney's 12 animation principles, timing windows for attacks/dodges, frame data, code examples.	<code>Claude Code skill marketplace</code>
ai-video-generation	claudeskills.info	Generate AI videos with 40+ models via inference.sh CLI. Useful for Year 3 AI layer.	<code>Claude Code skill marketplace</code>
remotion-render	skills.sh	Render videos from React/Remotion component code. Useful if we go React-based rendering.	<code>Claude Code skill marketplace</code>
claude-code-game-development	github.com/HermeticOrmus/claude-code-game-development	Game dev patterns: game loops, state	<code>Clone repo, reference patterns</code>

		management, input, collision, physics, animation, camera. All relevant patterns for our engine.
"Animate" Skill	claudemarketplaces.com	Create animated videos and motion graphics from natural language descriptions.

Claude Code skill marketplace

For Claude Code (Engineering Lead): Install **claudedesignskills** and **claude-code-game-development** before starting. The game development patterns (animation, camera, state management) map directly to our engine needs.

5. Codex CLI Skills for Animation/Graphics

Skill	Source	What It Does	Install
awesome-codex-skills	github.com/composio-community/awesome-codex-skills	14.4k+ stars. Curated list of 150+ practical Codex skills, subagents, MCP servers. Includes animation, video, and graphics skills.	Clone repo, browse skills
motion-skills	skillsllm.com	50 open-source skills that teach Codex to make motion graphics, animation & video — kinetic typography, data-viz, explainers.	Codex skill marketplace
svg-animations	github.com/topics/codex-skill	SVG animation skill for Codex. Good for generating vector-based stickman frames.	Codex skill marketplace
remotion-render (Codex)	skills.sh	Render Remotion videos from Codex. Same as Claude Code skill.	Codex skill marketplace

For Codex CLI (Core Engine Lead): Install **awesome-codex-skills** (reference the video/animation skills) and **motion-skills** for motion graphics patterns. The SVG animation skill is useful if we go with SVG-based rendering.

6. Competitive Landscape (What Exists Today)

Product	Type	Text-to-Stickman?	Price	Notes
Pivot Animator	Manual	No	Free	Most popular stickman tool. Manual frame-by-frame.
Stick Nodes	Manual	No	Free	Mobile stickman animator. Manual.
Vyond	Cartoon	No	\$649/yr	Polished animation, drag-drop templates.
Animaker	Cartoon	No	\$300/yr	2D animation, drag-drop.
AnimCreator	AI Stickman	Yes	Paid	animcreator.studio — closest competitor. Text-to-stickman, timeline, voice sync.
Stickman-Studio (OSS)	AI Pipeline	Yes	Free	Uses Gemini/Imagen. Not deterministic engine like ours.
StickFrame	Deterministic Engine	Yes	\$0-\$49/mo	Unique position. No AI needed. Pure deterministic engine.

StickFrame's Advantage: Every existing stickman tool is either (a) manual frame-by-frame, or (b) AI-based (expensive/unpredictable). A deterministic text-to-stickman engine that runs on CPU with zero API costs is a genuinely unique position. AnimCreator is the closest but is proprietary/paid.

7. Engineering Recommendations

For Claude Code (Engineering Lead)

- Use **pytweening** for all easing/tweening — `pip install pytweening`, import and done. 0 effort.
- Reference **The-StickMan/stickman.py** for joint rotation math (RotateableLine class)
- Reference **Axis** for keyframe animation approach
- Install **claude-designskills** — load animation/graphics skills
- Study **Stickman-Studio pipeline** for architecture inspiration (not code reuse)

For Codex CLI (Core Engine Lead)

- Evaluate **movis** as a potential rendering backend — test with Python 3.13
- Use **Pillow + subprocess ffmpeg** as the safe default
- Install **motion-skills** for motion graphics patterns
- Reference **Stickman-Studio/ffmpeg stitching** for video export code
- Keep it simple — the rendering loop is: draw frame → save PNG → ffmpeg stitch

Where We Must Build From Scratch

Unique code that has no existing library to copy from:

1. **Scene data model** (Scene, Character, Camera, Timeline) — custom to StickFrame
2. **Stickman animation keyframes** (walk/run/jump/attack/fall data) — custom authored
3. **Script-to-scene parser** (text input → structured scene commands) — unique to StickFrame
4. **Action timeline system** (character:hero:walk at t=2.0) — custom DSL

Everything else can leverage existing open-source libraries.

8. Link Index (All Resources)

Repos

- [Stickman-Studio](#) — AI stickman pipeline (reference)
- [The-StickMan](#) — Kivy stickman framework (math reference)

- [StickMotion](#) — Diffusion stickman motion
- [DanceStickFigure](#) — MediaPipe stickman
- [Axis](#) — Keyframed stickman animator
- [bone-animator](#) — Pivot clone
- [movis](#) — Python video editing engine (MIT)
- [SyncToon](#) — AI 2D animation from scripts
- [claudedesignskills](#) — Claude Code animation skills
- [claude-code-game-development](#) — Game dev patterns
- [awesome-codex-skills](#) — Curated Codex skills

Python Packages

- [pytweening](#) — Easing functions
- [movis](#) — Video editing engine
- [drawsvg](#) — SVG generation
- [ffmpeg-python](#) — FFmpeg bindings
- [PyAV](#) — Native FFmpeg interface
- [moviepy](#) — Video editing

Competitors & Market

- [AnimCreator](#) — Closest competitor (AI stickman, paid)
- [Pivot Animator](#) — Most used stickman tool
- [SVG Animate AI](#) — Online stick figure AI tool
- [Vidnoz](#) — Stick figure animator AI

Claude Code / Codex Skills

- [claudedesignskills](#) — 3D/WebGL/animation skills for Claude Code
- [Animation Principles Skill](#) — Disney principles for Claude Code
- [AI Video Generation Skill](#) — Video gen for Claude Code
- [Animate Skill](#) — Motion graphics skill
- [motion-skills](#) — Motion graphics for Codex
- [awesome-codex-skills](#) — 150+ Codex skills

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